Introduction to Robotics

Lecture 8: Trajectory Generation

Konstantinos Chatzilygeroudis - costashatz@upatras.gr

Department of Electrical and Computer Engineering
University of Patras

Template made by Panagiotis Papagiannopoulos

Trajectory Generation

- How can we generate a full trajectory (with timings) given an initial, $\boldsymbol{x}_s$ and target, $\boldsymbol{x}_g$, state?
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- What if $t > T$ or if I do not start from $t = 0$?

$$\boldsymbol{x}(t) = \boldsymbol{x}_s + \max(1, \frac{t - t_0}{T})(\boldsymbol{x}_g - \boldsymbol{x}_s)$$

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- Using the above, we model the path as $\boldsymbol{q}(t) = \boldsymbol{c}_1 t + \boldsymbol{c}_0$. Thus $\dot{\boldsymbol{q}}(t) = \boldsymbol{c}_1$!!! What can we do?
- Let's use higher order polynomial: $\boldsymbol{q}(t) = \boldsymbol{c}_2 t^2 + \boldsymbol{c}_1 t + \boldsymbol{c}_0$
- Now $\dot{\boldsymbol{q}}(t) = 2\boldsymbol{c}_2 t + \boldsymbol{c}_1$
- Much better!

Trajectory Generation (3)

$$\boldsymbol{q}(t) = \boldsymbol{c}_2 t^2 + \boldsymbol{c}_1 t + \boldsymbol{c}_0$$
$$\dot{\boldsymbol{q}}(t) = 2\boldsymbol{c}_2 t + \boldsymbol{c}_1$$

- What now? How can we compute the $\boldsymbol{c}_i$s?
- Given $\boldsymbol{x}_s = \begin{bmatrix} \boldsymbol{q}_s \\ \dot{\boldsymbol{q}}_s \end{bmatrix}$, $\boldsymbol{x}_g = \begin{bmatrix} \boldsymbol{q}_g \\ \dot{\boldsymbol{q}}_g \end{bmatrix}$ and a total time T:

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$$\boldsymbol{q}(T) = \boldsymbol{q}_g = \boldsymbol{c}_2 T^2 + \boldsymbol{c}_1 T + \boldsymbol{c}_0$$
$$= \boldsymbol{c}_2 T^2 + \dot{\boldsymbol{q}}_s T + \boldsymbol{q}_s \Rightarrow \boldsymbol{c}_2 = \frac{\boldsymbol{q}_g - \dot{\boldsymbol{q}}_s T - \boldsymbol{q}_s}{T^2}$$

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$$\begin{aligned} \boldsymbol{q}(T) = \boldsymbol{q}_g &= \boldsymbol{c}_2 T^2 + \boldsymbol{c}_1 T + \boldsymbol{c}_0 \\ &= \boldsymbol{c}_2 T^2 + \dot{\boldsymbol{q}}_s T + \boldsymbol{q}_s \Rightarrow \boldsymbol{c}_2 = \frac{\boldsymbol{q}_g - \dot{\boldsymbol{q}}_s T - \boldsymbol{q}_s}{T^2} \end{aligned}$$

- OR

$$\dot{\boldsymbol{q}}(T) = \dot{\boldsymbol{q}}_g = 2\boldsymbol{c}_2 T + \dot{\boldsymbol{q}}_s \Rightarrow \boldsymbol{c}_2 = \frac{\dot{\boldsymbol{q}}_g - \dot{\boldsymbol{q}}_s}{2T}$$

Cubic Splines

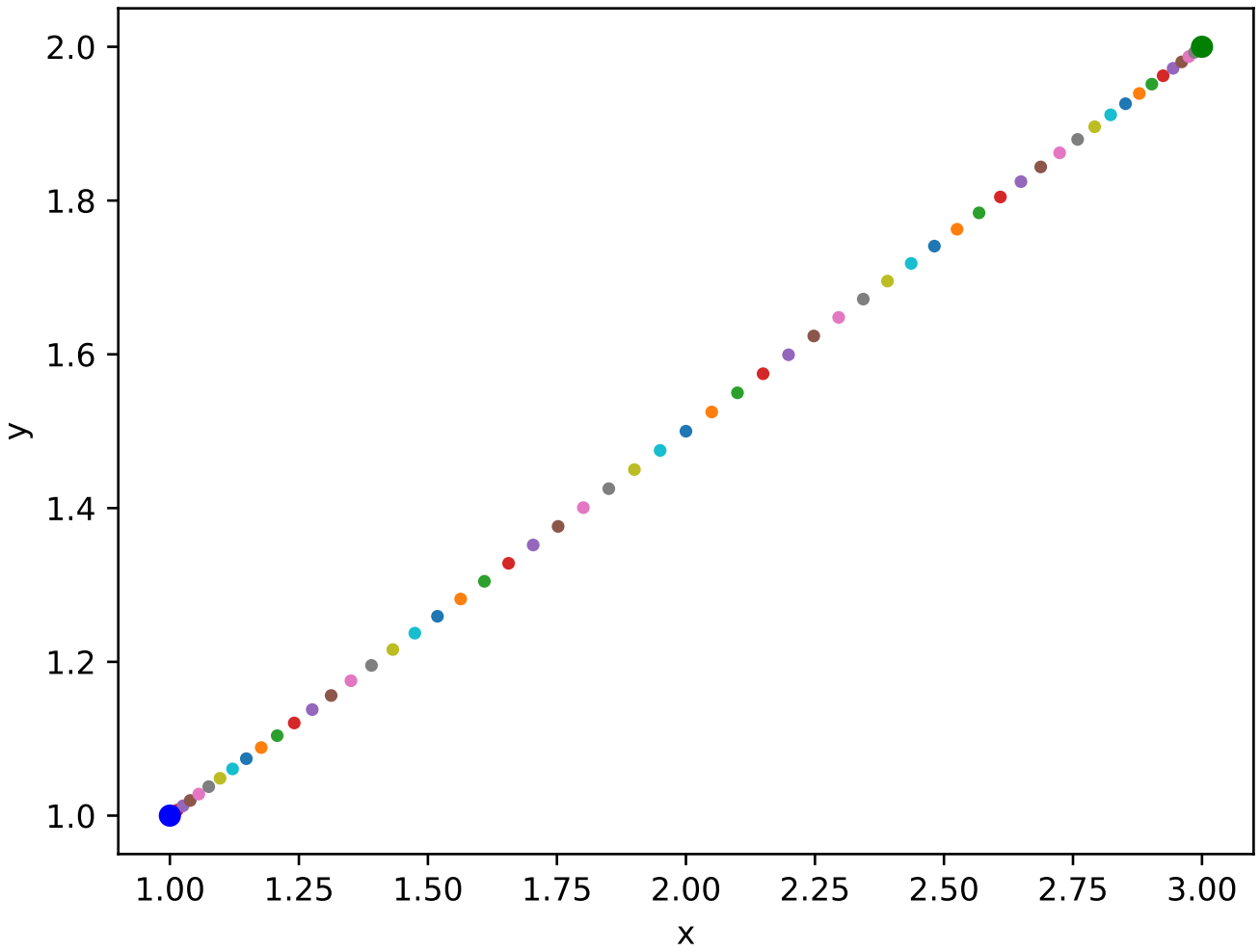
- Even-ordered polynomials are not good! We are always left with overparameterized systems!
- Let's do 3rd order:

$$\boldsymbol{q}(t) = \boldsymbol{c}_3 t^3 + \boldsymbol{c}_2 t^2 + \boldsymbol{c}_1 t + \boldsymbol{c}_0$$

- And:

$$\begin{aligned}
\boldsymbol{c}_0 &= \boldsymbol{q}_s \\
\boldsymbol{c}_1 &= \dot{\boldsymbol{q}}_s \\
\boldsymbol{c}_2 &= \frac{3\boldsymbol{q}_g}{T^2} - \frac{3\boldsymbol{q}_s}{T^2} - \frac{2\dot{\boldsymbol{q}}_s}{T} - \frac{\dot{\boldsymbol{q}}_g}{T} \\
\boldsymbol{c}_3 &= -\frac{2\boldsymbol{q}_g}{T^3} + \frac{2\boldsymbol{q}_s}{T^3} + \frac{\dot{\boldsymbol{q}}_s}{T^2} + \frac{\dot{\boldsymbol{q}}_g}{T^2}
\end{aligned}$$

Splines - Code Example



Trajectory Generation with SO(3)?

- What happens when we are in SO(3)?

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- Let's first see the straight line path. We **CANNOT** do:

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- First we need to replace $\boldsymbol{x}$ with $\boldsymbol{R}$. Now what?

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- Easy! We just interpolate!

$$\begin{aligned}\boldsymbol{R}(t) &= \boldsymbol{R}_s(\boldsymbol{R}_s^T\boldsymbol{R}_g)^t \\ &= \boldsymbol{R}_s \exp(t\log(\boldsymbol{R}_s^T\boldsymbol{R}_g))\end{aligned}$$

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- We can use the equations as before, almost:

$$\boldsymbol{R}(t) = \exp(\hat{\boldsymbol{c}}_3 t^3 + \hat{\boldsymbol{c}}_2 t^2 + \hat{\boldsymbol{c}}_1 t + \hat{\boldsymbol{c}}_0)$$
$$\dot{\boldsymbol{R}}(t) = (3\hat{\boldsymbol{c}}_3 t^2 + 2\hat{\boldsymbol{c}}_2 t + \hat{\boldsymbol{c}}_1)\boldsymbol{R}(t)$$

- Thus if we start at $t = 0$ from $(\boldsymbol{R}_s, \boldsymbol{\omega}_s)$ and we want to arrive at $t = T$ to $(\boldsymbol{R}_g, \boldsymbol{\omega}_g)$, we have:

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- Thus if we start at $t = 0$ from $(\boldsymbol{R}_s, \boldsymbol{\omega}_s)$ and we want to arrive at $t = T$ to $(\boldsymbol{R}_g, \boldsymbol{\omega}_g)$, we have:

$$\boldsymbol{R}(0) = \exp(\hat{\boldsymbol{c}}_0) = \boldsymbol{R}_s \Rightarrow \boldsymbol{c}_0 = \widehat{\log(\boldsymbol{R}_s)} = \boldsymbol{\phi}_s$$
$$\dot{\boldsymbol{R}}(0) = \hat{\boldsymbol{c}}_1 \boldsymbol{R}(0) \Rightarrow \hat{\boldsymbol{\omega}}_s \boldsymbol{R}_s = \hat{\boldsymbol{c}}_1 \boldsymbol{R}_s \Rightarrow \boldsymbol{c}_1 = \boldsymbol{\omega}_s$$
$$\boldsymbol{c}_2 = \frac{3\boldsymbol{\phi}_g}{T^2} - \frac{3\boldsymbol{\phi}_s}{T^2} - \frac{2\boldsymbol{\omega}_s}{T} - \frac{\boldsymbol{\omega}_g}{T}$$
$$\boldsymbol{c}_3 = -\frac{2\boldsymbol{\phi}_g}{T^3} + \frac{2\boldsymbol{\phi}_s}{T^3} + \frac{\boldsymbol{\omega}_s}{T^2} + \frac{\boldsymbol{\omega}_g}{T^2}$$

where $\boldsymbol{R}_s = \exp(\hat{\boldsymbol{\phi}}_s), \boldsymbol{R}_g = \exp(\hat{\boldsymbol{\phi}}_g)$.

Cubic Splines with SE(3)

- We do the same as before:

$$\boldsymbol{T}(t) = \exp(\hat{\boldsymbol{c}}_3 t^3 + \hat{\boldsymbol{c}}_2 t^2 + \hat{\boldsymbol{c}}_1 t + \hat{\boldsymbol{c}}_0)$$
$$\dot{\boldsymbol{T}}(t) = (3\hat{\boldsymbol{c}}_3 t^2 + 2\hat{\boldsymbol{c}}_2 t + \hat{\boldsymbol{c}}_1)\boldsymbol{T}(t)$$

- Thus if we start at $t = 0$ from $(\boldsymbol{T}_s, \mathcal{V}_s)$ and we want to arrive at $t = T$ to $(\boldsymbol{T}_g, \mathcal{V}_g)$, we have:

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- Thus if we start at $t = 0$ from $(\boldsymbol{T}_s, \mathcal{V}_s)$ and we want to arrive at $t = T$ to $(\boldsymbol{T}_g, \mathcal{V}_g)$, we have:

$$\boldsymbol{T}(0) = \exp(\hat{\boldsymbol{c}}_0) = \boldsymbol{T}_s \Rightarrow \boldsymbol{c}_0 = \widehat{\log(\boldsymbol{T}_s)} = \boldsymbol{\tau}_s$$
$$\dot{\boldsymbol{T}}(0) = \hat{\boldsymbol{c}}_1 \boldsymbol{T}(0) \Rightarrow [\mathcal{V}_s]\boldsymbol{T}_s = \hat{\boldsymbol{c}}_1 \boldsymbol{T}_s \Rightarrow \boldsymbol{c}_1 = \mathcal{V}_s$$
$$\boldsymbol{c}_2 = \frac{3\boldsymbol{\tau}_g}{T^2} - \frac{3\boldsymbol{\tau}_s}{T^2} - \frac{2\mathcal{V}_s}{T} - \frac{\mathcal{V}_g}{T}$$
$$\boldsymbol{c}_3 = -\frac{2\boldsymbol{\tau}_g}{T^3} + \frac{2\boldsymbol{\tau}_s}{T^3} + \frac{\mathcal{V}_s}{T^2} + \frac{\mathcal{V}_g}{T^2}$$

where $\boldsymbol{T}_s = \exp(\hat{\boldsymbol{\tau}}_s)$, $\boldsymbol{T}_g = \exp(\hat{\boldsymbol{\tau}}_g)$.

Bibliography

Chapter 9 from **Modern Robotics: Mechanics, Planning, and Control**, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press. ebook

Thank you

- **Any Questions?**
- **Office Hours**:
  - **Tue-Thu** *(10:00-12:00)*
  - 24/7 by email (`costashatz@upatras.gr`, subject: *ECE_IR_AM*)
- **Material and Announcements**

Laboratory of Automation & Robotics